

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2361

Amargo 138 kV
December 04, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2361. The Interconnection Customer proposes a 82.6 MW solar photovoltaic facility interconnecting at the Amargo 138 kV substation in the Virginia area. The request was received on November 28, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$646,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: NEW LONDON - THORNHILL 138kV
Contingency: DESERT OUTPOST SUBSTATION - MONEL 138kV
Loading: 106.0% (Rating: 242 MVA)
DFAX: 5.4%
Pre-project loading: 104.2%
- Facility: THORNHILL - CARSON ICE COGEN 138kV Line
Contingency: MONEL - DESERT OUTPOST 138kV
Loading: 101.6% (Rating: 234 MVA)
DFAX: 5.4%
Pre-project loading: 99.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEW LONDON SUBSTATION - THORNHILL 138kV	Reconductor line	\$4,714,000	8% (\$380,000)
THORNHILL - CARSON ICE COGEN 138kV Line	Reconductor line	\$3,305,000	8% (\$266,000)
	Total allocated to project		\$646,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.